

2	Define concurrency. Explain the need for concurrent execution. List and describe the various problems encountered due to concurrent execution with relevant examples. Definition : Concurrency refers to the simultaneous execution of two or more transactions in a database system such that their operations are interleaved over time, rather than executed one after another. Need for Concurrent Execution Concurrent execution of transactions is necessary for the following reasons: Improved Throughput Better Resource Utilization. Reduced Waiting Time Support for Multiuser Environment Lost Update Problem Description: Occurs when two transactions update the same data item and one update is lost due to overwriting by another transaction. Example: • T₁ reads account balance = 1000 • T₂ reads account balance = 1000 • T₁ updates balance to 900 • T₂ updates balance to 1100 Final balance becomes 1100, and T₁'s update is lost. Dirty Read (Uncommitted Dependency) Description: Occurs when a transaction reads data written by another transaction that has not yet committed. Example: • T₁ updates balance from 1000 to 800 (not committed) • T₂ reads balance as 800 • T₁ rolls back T₂ has read invalid data.	10
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	Unrepeatable Read Description: Occurs when a transaction reads the same data item twice and gets different values due to modification by another committed transaction. Example: T_1 reads balance = 1000 T_2 updates balance to 1200 and commits T_1 reads balance again $\rightarrow$ 1200 Same read produces different results. Incorrect Summary Problem Description: Occurs when a transaction computes an aggregate function while another transaction updates some of the involved data items. Example: • T_1 calculates total balance of all accounts • T_2 transfers money between two accounts during calculation Resulting sum is inconsistent.	
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